

Exploration or Numerics? A Pre-Registered, Optimiser-Controlled Study of Greedy Bayesian Optimisation and LogEI on De Ath’s Suite

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Abstract

Two influential results about Expected Improvement (EI), a central acquisition function in Bayesian optimisation, point in opposite directions. [De Ath et al. \(2021\)](#) show that near-greedy exploitation matches or beats EI on a synthetic suite, locating the difficulty in EI’s explore–exploit decision rule; [Ament et al. \(2023\)](#) show that a numerically repaired EI, LogEI, improves on raw EI, locating it instead in EI’s numerics. Whether De Ath’s greedy-beats-EI result was partly an artefact of numerically broken EI has not been tested. The two results sit on a never-crossed axis: De Ath used a gradient-free acquisition optimiser, while the field has largely standardised on gradient-based optimisation, under which alone the EI pathology has been analysed. We pre-register and run a 1650-cell, optimiser-controlled experiment on De Ath’s suite, adding a faithful gradient-free (NSGA-II) optimiser beside the standard gradient-based one, over a reproduction of his published medians grounded by an author-confirmed audit. Repairing EI’s numerics does not help here: LogEI beats neither raw EI nor De Ath’s eps-greedy method, so we find no support for a broken-EI explanation. Pure greedy collapses to the worst arm overall, yet under NSGA-II the best of the planned arms is an eps-greedy method. In high dimensions greedy beats EI and LogEI but meets the per-problem criterion on only 2 of 4 problems: that reproduction fails, and the design principle that a little exploration suffices is supported descriptively, not by the planned test. The EI gradient pathology is real and probe-confirmed but yields no performance advantage here, and the two optimisers largely preserve the arm ordering. On this suite the two findings are compatible: what matters is that some exploration is present, not which acquisition supplies it. None of the three criteria we fixed in advance came out positive; each resolved negative or at a boundary.

1 Introduction

Bayesian optimisation (BO) chooses where to evaluate an expensive black-box function by maximising an acquisition function over a Gaussian-process surrogate ([Frazier, 2018](#)). EI is among the most widely used acquisition functions in practice ([Jones et al., 1998](#); [Frazier, 2018](#)). Two recent results disagree about it at their strong readings. [De Ath et al. \(2021\)](#) argue that principled acquisitions over-explore: a near-greedy strategy, which selects the posterior-mean maximiser with a small random-exploration probability, matches or beats EI and UCB, increasingly so as dimension grows. [Ament et al. \(2023\)](#) argue the opposite about the same acquisition function: EI is sound but numerically broken, and a log-space reformulation (LogEI) that repairs the numerics recovers strong performance. A practitioner who picks an acquisition function today gets contradictory advice from these two lines of work. One line says the default over-explores and near-greedy selection does better, while the other says the default is sound once its numerics are repaired. Which reading is right determines where improvement effort should go, into exploration schedules or into numerical implementation. The disagreement also illustrates a problem that reaches beyond BO. Two influential empirical results can attribute the same observed behaviour to different mechanisms, one to the decision rule and one to its numerical implementation, and remain uncontested for years because no experiment isolates the difference. This paper puts one such case to a controlled test, with the analysis plan frozen in advance.

The two readings are in tension: if De Ath’s result were partly an artefact of broken numerics, repairing the numerics should reverse it. LogEI should then beat the near-greedy strategy. Ament et al. (2023) invite this reading by attributing EI’s inconsistent and often sub-optimal reported performance across the literature to optimisation failure rather than to the acquisition itself. We refer to this hypothesis as the broken-EI explanation of De Ath, and we test it directly.

The question has stayed open because the two results live on different axes that have not been crossed. De Ath optimised every acquisition with a gradient-free evolutionary optimiser. Much of the field, and Ament in particular, standardised on gradient-based multi-start L-BFGS-B (Wilson et al., 2018; Balandat et al., 2020), and Ament analyses the pathology only under gradient-based optimisation. No prior study varies the acquisition-*optimiser* type while holding the problem suite and the EI-versus-LogEI contrast fixed. Adjacent work approaches but does not occupy this intersection. Wilson et al. (2018) study acquisition maximisation itself and compare gradient-based with random optimisers, but they predate LogEI and do not cross the optimiser axis with the EI-versus-LogEI contrast or De Ath’s suite. Georgiou et al. (2025) vary the acquisition optimiser between deterministic-global and local or multi-start solvers and find that suboptimal optimisation can be preferable, yet they study LCB and do not engage the EI pathology. Rehbach et al. (2020) independently report that predicted-value exploitation beats EI for $d \geq 5$, but only under gradient-free optimisation (differential evolution). Xie et al. (2024) advance cost-aware acquisition design, still within gradient-based optimisation.

We test what remains open on De Ath’s own suite with a pre-registered, optimiser-controlled experiment. Repairing EI’s numerics does not help: LogEI beats neither raw EI nor De Ath’s eps-greedy method, so we find no support for the broken-EI explanation, and the gap is far too small to reverse the greedy result. Pure exploitation collapses to the worst arm. Under NSGA-II, however, the best pre-registered arm is an eps-greedy method, our documented variant of De Ath’s eps-PF, so his design principle that a little exploration suffices holds even as the zero-exploration limit fails. The advantage that LogEI shows over EI under gradient-based optimisation does not shrink under gradient-free optimisation, because there is little advantage to shrink. These outcomes confirm a boundary that Ament et al. (2023) state in their own discussion rather than contradicting them.

A single mechanism is consistent with the optimiser invariance we observe. The EI pathology has two parts: acquisition values that underflow to floating-point zero, and gradients that vanish. A gradient-free optimiser bypasses the second part but not the first, since a region where EI is exactly zero is unrankable by any optimiser. The claim that De Ath’s EI was broken is thus partly true and partly false. In principle value-underflow could have affected his gradient-free runs, and our probe does find it at $d = 1, 2$, whereas gradient-vanishing never applied to a gradient-free optimiser. Value-underflow is a property of the acquisition surface, not of the optimiser, and it is negligible at $d \geq 6$ (below 3×10^{-5}); this is consistent with the closely matching orderings we observe.

This paper makes three contributions. The first consists of pre-registered boundary findings which, taken together, address the De Ath–Ament tension on this suite. These are a bounded null for the LogEI advantage; a corroboration of eps-greedy, along with greedy competitiveness in high dimension and a collapse of pure greedy; and a broad preservation of the arm ordering across optimisers. The second is an author-confirmed reproduction audit of De Ath et al. (2021) that grounds the negative results. The third is a pre-registration methodology (Nosek et al., 2018), with a plan frozen and hashed before any production data (as defined in Section 4.1), that lets the negative results be read at face value. Part 1 is the foundation; the main findings are in Sections 4.3–4.7.

2 Background and related work

De Ath et al. (2021). The reproduction target proposes eps-greedy acquisition. With probability $1 - \epsilon$ the next point is the posterior-mean maximiser, $\arg \max_x \mu(x)$, which is pure exploitation; with probability ϵ an exploratory point is drawn, either uniformly (eps-RS) or from the non-dominated set of the posterior (μ, σ) trade-off (eps-PF). De Ath reports that exploitation-heavy strategies are competitive, and increasingly so with dimension, optimising every acquisition with NSGA-II at a budget of $5000d$ evaluations (population $100d$, 50 generations). Their proposed methods use $\epsilon = 0.1$; the exploit component is the $\epsilon \rightarrow 0$ limit.

Ament et al. (2023). LogEI repairs two numerical pathologies of EI that arise under gradient-based optimisation: acquisition values that are numerically zero over large regions, and gradients that vanish in those regions (their Theorem 1). Their acquisition optimiser is multi-start L-BFGS-B (their §5), and they focus on gradient-based optimisation by design (their §2.3). In their own discussion (their §6), Ament places bounds on the advantage. On their account the gain grows with dimension because high objective values concentrate in the search space, not because of dimensionality as such, and where the surrogate’s predictions or uncertainties are poor, LogEI need not outperform EI. A support-vector-machine problem with high ambient but low intrinsic dimension shows no significant LogEI gain (their Fig. 6). Our results systematise this boundary rather than contesting it.

Adjacent optimiser and acquisition studies. Georgiou et al. (2025) compare a deterministic-global solver (MAiNGO) against local and multi-start L-BFGS-B and find that suboptimal optimisation of a poorly chosen acquisition can be preferable to its optimum; their study uses LCB and does not touch the EI gradient pathology. Rehbach et al. (2020) find predicted-value exploitation beats EI for $d \geq 5$ on the BBOB suite using differential evolution, an independent gradient-free confirmation of greedy competitiveness that we test for robustness across optimiser family. Xie et al. (2024) develop a cost-aware acquisition from the Pandora’s Box Gittins index and evaluate it within gradient-based optimisation.

Position of this study. No prior work tests greedy-versus-principled selection and raw-EI-versus-LogEI together, on De Ath’s suite, across the acquisition-optimiser-type axis. That intersection is exactly where this study sits. A gradient-free optimiser separates value-underflow from gradient-vanishing operationally, because it is immune to the second pathology but not the first; Ament et al. (2023) analyse the pathology only under gradient-based optimisation and do not perform this separation.

3 Part 1: reproduction and author-confirmed audit

This part establishes that our stack reproduces De Ath and that our reading of his materials is verifiable. It is the basis for trusting the negative results in Part 2. The author correspondence below supports Part 1 only; no Part 2 finding depends on it.

3.1 Exact reproduction

Before any new statistic is computed, we reproduce every median-regret cell of De Ath’s Table 2 to three significant figures from his published runs. The validation gate passes on 90/90 cells, and one representative problem appears in Table 4 (Appendix A). In De Ath’s labels, eFront and eRandom are the eps-PF and eps-RS eps-greedy methods and Exploit is pure greedy, matching the arm names used in Part 2. A cross-problem Friedman test on the reproduced medians confirms strategy separation (10 problems, 9 methods, $\chi^2(8) = 50.7$, $p = 3.0 \times 10^{-8}$), with the eps-greedy methods at the best average ranks.

We also re-execute his published pipeline inside his Docker image: 88 fresh runs (Branin and logGSobol $\times$ four methods $\times$ 11 runs). No cell differs from his published 51-run distribution: every paired sign test gives $p \geq 0.22$, the paired median log-regret differences satisfy $|\Delta \log| \leq 0.90$, and the smallest Mann–Whitney p -value, 0.10 (logGSobol-EI), rises to 0.81 under Holm correction. The reproduction is distributional rather than bitwise. Per-run trajectories diverge across re-runs from non-deterministic parallel reductions in the GP fit, a property we observe in both De Ath’s stack and our own.

3.2 Four text-versus-code discrepancies

The audit found four discrepancies between De Ath’s paper text and his released code, summarised in Table 1. None changes the best-median method on any problem. F1, the one with the largest scope, shifts the exact logSixHumpCamel regret values, because the stored reference optimum was evaluated at rounded coordinates. F3 touches nothing beyond the equivalence annotations on WangFreitas and logRosenbrock, where the code runs a two-sided test in place of the one-sided test described in the text.

Table 1: The four author-confirmed text-versus-code discrepancies in De Ath’s released code. The numbering skips F4, a separate observation about Ament’s optimiser-restart protocol (sixteen versus twenty restarts within one section) that is not a De Ath discrepancy. All four here are limited in scope and leave the median-performance conclusions unchanged.

ID	Problems affected	Discrepancy
F1	logSixHumpCamel	Regret is $ y - y_{\text{opt}} $ before the cumulative minimum, with y_{opt} at rounded optimum coordinates; changes exact regret values and one equivalence annotation, not the best-median method.
F2	all (tabulated MAD)	The tabulated MAD is SciPy’s scaled MAD ($\times 1.4826$); the text states the unscaled definition.
F3	WangFreitas, logRosenbrock	The code calls a two-sided Wilcoxon test; the text states one-sided. Changes equivalence annotations, not medians.
F5	gSobol	The supplementary gSobol expression does not reproduce the implemented function, which matches the González local-penalisation form (González et al., 2016).

3.3 Author confirmation

George De Ath confirmed F1, F2, F3, and F5 as real text-versus-code discrepancies rather than undocumented choices, and characterised them as limited in scope and not materially affecting the paper’s conclusions (G. De Ath, personal communication, 2026; permission to report granted).¹ He noted that the best-median methods are unchanged and that EI, PI, and UCB remain not-best-median on any synthetic problem. He reaffirmed that the exploration-matters conclusion stands. A second reply added a clarification: the two regret orderings are equivalent when y_{opt} is the true optimum, so F1’s materiality concerns the rounded reference optimum; the ordering is unaffected. The reply also traced the logSixHumpCamel annotation change to F1 and the WangFreitas and logRosenbrock changes to F3. Throughout the exchange he maintained the greedy-is-good claim.

4 Part 2: a pre-registered optimiser-axis study

The Part-2 pre-registered verdicts stand on the 1650-cell NSGA-II matrix and its paired gradient-based counterpart alone; the post-hoc additions of Section 4.7 are declared separately and enter no verdict.

4.1 Pre-registration and integrity chain

Before any production data existed, we froze the full analysis plan (hypotheses, estimands, decision criteria, and the probe protocol) and recorded it under a git tag. The tags, commits, and content hashes behind every step are listed in Appendix B and the reproducibility statement. The post-data record is an append-only amendment that records the hashes of the analysis script and the data archive; it alters no hypothesis or criterion. The confirmatory analysis is separated from the frozen plan by construction: it imports the prior analysis’s regret helpers read-only and selects no test, threshold, or aggregation from the data. The separation is between the plan and its executor; no independent human reviewer was involved. Production data here means the 1650 NSGA-II cells and their probe streams. The paired gradient-based matrices predate the freeze and had been analysed, so the H3 direction is a conditional prediction about not-yet-existing gradient-free data given the observed gradient-based side; Appendix B reconstructs the event order. The same discipline produced the Part-1 reproduction freeze and the gradient-matrix analysis plan, the latter committed before its data existed.

¹The correspondence is available upon request.

4.2 Setup

We add five NSGA-II acquisition arms to the frozen gradient-based matrix, changing only the acquisition optimiser: EI, LogEI, UCB, Exploit (pure greedy, $\arg \max \mu$), and eps-PF. The gradient-based arms optimise their acquisitions with BoTorch multi-start L-BFGS-B (ten restarts, 64 raw samples) (Balandat et al., 2020). The NSGA-II arms build the same acquisition objects and optimise them with a pymoo genetic algorithm (Blank and Deb, 2020; Deb et al., 2002) at De Ath’s settings: population $100d$, 50 generations, SBX crossover ($p = 0.8$, $\eta = 20$), polynomial mutation ($p = 1/d$, $\eta = 20$). The two optimiser regimes run at their ecologically valid settings, without matching the acquisition-evaluation budget across regimes: the NSGA-II arms follow De Ath’s published configuration (5000 d acquisition evaluations per step) and the gradient-based arms follow standard BoTorch practice. The comparison is therefore between two field-standard regimes. The problem suite is De Ath’s ten synthetic problems together with Ackley ($d = 8$); the latter is not one of De Ath’s problems and is added at pre-registration so the dimension axis spans five distinct values, $\{1, 2, 6, 8, 10\}$, against De Ath’s four. Part 2 uses this repository’s problem labels; Table 10 maps them to De Ath’s names. Each problem is run for 30 seeds and $T = 250$ evaluations. This gives $5 \times 11 \times 30 = 1650$ NSGA-II cells, paired against 1650 gradient-based cells drawn from an earlier 2400-cell core matrix (the ten De Ath problems) and a 720-cell extension matrix, which includes Ackley, so 3300 paired cells in total. Surrogate fits use GPyTorch (Gardner et al., 2018), and the analyses use SciPy (Virtanen et al., 2020) and statsmodels (Seabold and Perktold, 2010). The pairing of each NSGA-II cell with its gradient-based counterpart rests on a byte-identical initial design. The injected design is identical by construction, because before the acquisition step both optimisers either load the same De Ath design or draw the same seeded Sobol (Sobol’, 1967) fallback. We confirmed a maximum absolute difference of 0 on 22 spot-checked cells including the Sobol-fallback problem. Across the 3300 paired cells the precise final regret is finite everywhere, with 0/3300 negative-regret clips. A frozen mechanism probe evaluates 4096 Sobol points each iteration to record value-underflow and gradient health.

4.3 H1: the dimensional trend is present but underpowered

The directional claim that pure greedy’s advantage over EI grows with dimension holds in sign under both optimisers but does not meet the pre-registered significance criterion. Within each optimiser we aggregate to a per-problem advantage A_p and regress it on $\log d$ across the 11 problems with HC3-robust standard errors. The slope is positive under both optimisers, yet the robust p -value clears 0.05 under neither (Table 3). The nonparametric sensitivity agrees in direction and reaches significance under the gradient-based optimiser only (Spearman $\rho = 0.71$, $p = 0.01$). H1 is not confirmed. The regression rests on five distinct dimensions over eleven problems, so this is the low-power outcome anticipated in the pre-registration.

4.4 H2a: a high-dimensional reproduction inside an aggregate failure

By the pre-registered criteria, evaluated on De Ath’s ten synthetic problems, the NSGA-II reproduction of his strategy ordering fails. Among the three anchor strategies, the equal-problem-weight mean log-regret orders them $UCB < EI < \text{Exploit}$, which places pure greedy worst. The rank correlation with De Ath’s ordering is -0.50 , below the 0.5 threshold. De Ath’s EI-versus-UCB sub-order is not crisply specified; the frozen reference fixes Exploit–UCB–EI, the alternative implied by his published average ranks gives -1.0 , and both admissible references fail. Pure greedy ranks at or above EI on 2 of the 4 high-dimensional De Ath problems, short of a majority. This rank correlation is over three strategies and can take only the values $\{\pm 1, \pm 0.5\}$, so it has low resolution by construction; we execute the criterion as pre-registered and set it beside the decomposition below.

The failure is an artefact of all-problem aggregation rather than a high-dimensional contradiction of De Ath, and the per-arm means make this explicit (Table 2, Figure 1). In high dimensions pure greedy beats both EI and LogEI under both optimisers; the worst-overall ranking is produced entirely by its low-dimensional collapse, where its mean log-regret over the low-dimensional problems is -2.20 against EI’s -4.67 under gradient-based optimisation. The decomposition therefore shows pure greedy to be competitive in high dimensions, where De Ath locates his claim, even though it is worst on the all-problem mean that the pre-registered criterion uses. The 2-of-4 count is that per-problem criterion and is what fails; we draw no clean high-dimensional win from it.

Table 2: Mean log precise regret M_a per arm and optimiser (lower is better), over De Ath’s ten problems and over their low- ($d \leq 2$) and high-dimensional ($d \geq 6$) subsets; Ackley ($d = 8$) is excluded to match De Ath’s suite, whereas H1 and H3 use all eleven problems. Pure greedy (Exploit) is worst overall but beats EI and LogEI in high dimensions; eps-PF is best; LogEI does not improve on EI. The best value in each column is in bold, including ties at the displayed precision. Per-cell seed-level dispersion is given in Tables 6–7.

Arm	All 10		High- d (≥ 6)		Low- d (≤ 2)	
	grad	nsga	grad	nsga	grad	nsga
eps-PF	−3.13	−3.50	−0.47	−0.43	−4.91	−5.54
UCB	−3.23	−3.24	−0.71	−0.71	−4.91	−4.92
EI	−2.83	−2.70	−0.06	−0.09	−4.67	−4.44
LogEI	−2.74	−2.58	−0.20	−0.10	−4.42	−4.23
Exploit	−1.45	−1.62	−0.32	−0.33	−2.20	−2.48

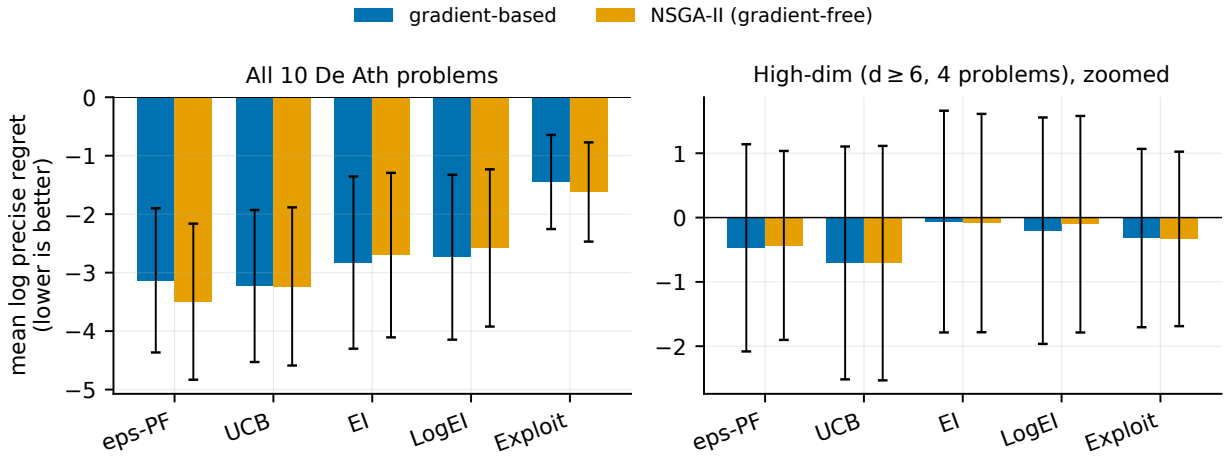


Figure 1: Mean log precise regret per arm under each optimiser, over De Ath’s ten problems (left) and their high-dimensional subset (right); whiskers give the problem-level standard error of each mean, the cross-problem spread that also drives the bootstrap intervals. Pure greedy (Exploit) is worst overall but moves up in high dimensions; eps-PF is best in point estimate; LogEI tracks EI; the gradient-based and gradient-free bars are close throughout. The right panel is zoomed (independent vertical scale); Appendix D shows the shared-axis version.

De Ath’s eps-greedy design, in contrast, is corroborated. Our eps-PF arm is a documented variant of his eFront: like his, it evolves the (mean, sigma) trade-off front with NSGA-II over the GP. It diverges in two ways: its exploit branch searches the full domain for the posterior-mean maximiser where he selects the best-mean front member, and its front is the evolved population’s non-dominated set where he archives every evaluated point. It is the single best pre-registered arm under NSGA-II, at a mean log-regret of -3.50 against the next-best UCB at -3.24 . This ordering is a point estimate. The same problem-level bootstrap used for the LogEI null puts the eps-PF-versus-UCB contrast at $+0.26$ with a 95% interval of $[-0.46, +0.93]$. The lead over UCB therefore lies inside problem-level noise, and the corroboration is descriptive, not confirmatory, in line with its pre-registered designation as the descriptive hypothesis H2b. The exploit component of eps-PF is exactly the pure-greedy arm that collapses, so the data support De Ath’s design principle that a little exploration suffices while bounding the zero-exploration limit, which fails on this stack. A post-hoc arm implementing his front-construction mechanics, up to documented divergences, strengthens this further: it is better still overall, though in high dimensions it too remains behind UCB (Section 4.7).

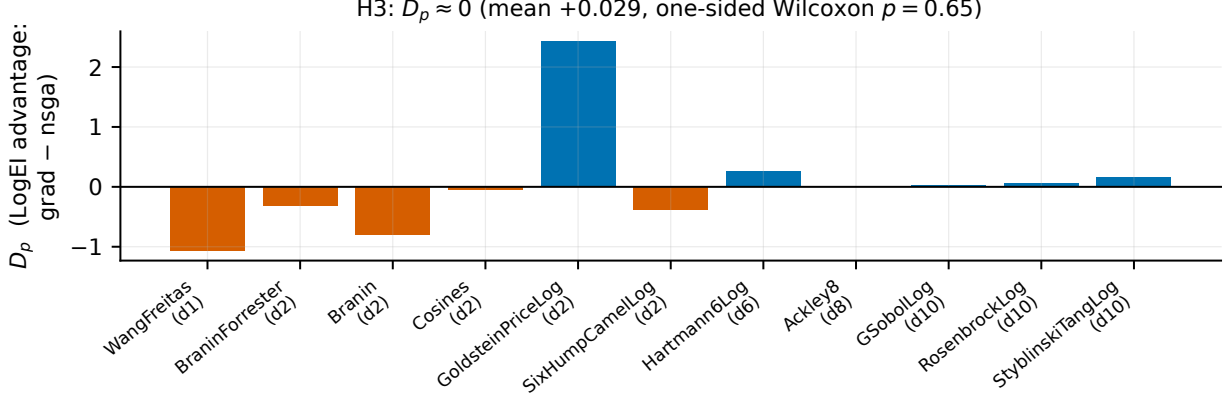


Figure 2: Per-problem H3 contrast D_p (LogEI advantage under gradient-based minus gradient-free optimisation) across all eleven problems, five of them with $d \geq 6$. The contrast is centred near zero; the pre-registered one-sided test is not significant.

4.5 H3: the LogEI advantage does not shrink

The pre-registered hypothesis that the EI-to-LogEI advantage shrinks under the gradient-free optimiser is rejected. We form the per-problem contrast D_p between the LogEI advantage under gradient-based and gradient-free optimisation, and test $D_p > 0$ across the 11 problems with a one-sided Wilcoxon signed-rank test. The contrast is centred near zero (mean +0.03, six of eleven problems positive) and the test is not significant ($W = 29$, $p = 0.65$). The paired t -test and sign test agree, and the seed-level mixed model is not significant in the predicted direction ($p = 0.46$). A post-hoc leave-one-out check shows the near-zero mean is itself outlier-dependent. Dropping GoldsteinPriceLog, the one problem with a large gradient-side LogEI advantage, moves the mean contrast to -0.21 (five of ten positive; one-sided Wilcoxon $p = 0.84$), further from the predicted direction: the rejection is not carried by that problem. Figure 2 shows the contrast per problem.

There is little LogEI advantage to shrink. LogEI’s mean advantage over EI is -0.08 under gradient-based and -0.11 under gradient-free optimisation across the eleven-problem suite, so LogEI does not improve on EI, and over De Ath’s ten problems it trails eps-PF by 0.92 under NSGA-II. A post-hoc problem-level bootstrap (Appendix E) brackets this. Over De Ath’s ten problems under NSGA-II the advantage is -0.12 with a 95% percentile interval of $[-0.30, +0.04]$, small against the 0.92 gap to eps-PF; the bootstrap resamples only ten problems, too few for calibrated tail endpoints. Under the gradient-based optimiser the interval is wide ($[-0.58, +0.50]$), so the null is tighter on the gradient-free side and loose on the gradient-based side. The gradient-based interval is widened by a single problem, GoldsteinPriceLog ($d = 2$), which shows by far the largest gradient-side LogEI advantage ($+2.13$, against -0.29 under NSGA-II). Repairing EI’s numerics therefore does not translate into BO performance on this suite. The result is a bound, not a refutation: whatever LogEI advantage exists is too small to reverse the greedy result in the gradient-free regime where that result lives, and neither optimiser shows support for the broken-EI explanation.

The probe shows why the mechanism has no effect here (Figure 3). EI gradients vanish over much of the domain at low dimension, as Ament predicts. The other pathology, value-underflow, is the part a gradient-free optimiser cannot escape, and the fraction of probe points where raw EI is exactly zero falls from 0.22 at $d = 1$ to below 3×10^{-5} by $d = 6$ and to 0 at $d = 10$. The binary value-versus-gradient disambiguation therefore does not cleanly resolve, precisely because the underlying EI-to-LogEI gap is small. The null still carries information: under gradient-free optimisation LogEI’s gradient benefit is irrelevant by construction, but its value-underflow mitigation could have helped at low dimension and does not. This is the operational form of Ament’s own caveat that better acquisition values need not yield better BO performance. The probe comparison across optimisers is descriptive, since the gradient-based probe samples 64 points and the gradient-free probe samples 4096.

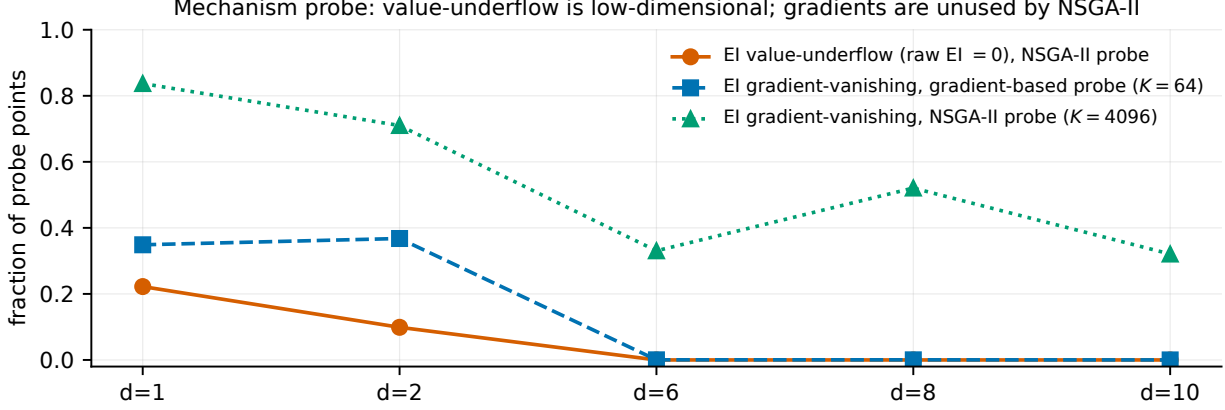


Figure 3: Mechanism probe for EI by dimension. Value-underflow (raw EI exactly zero) is a low-dimensional phenomenon: the fraction falls below 3×10^{-5} by $d = 6$ and to zero at $d = 10$. Gradient-vanishing is present but is not used by the gradient-free optimiser; the gradient-based and gradient-free curves differ because they sample 64 and 4096 probe points respectively. The NSGA-II gradient curve is shown for completeness; NSGA-II does not consume gradients.

Table 3: Pre-registered verdicts, reported in whichever direction they resolved.

Hypothesis	Pre-registered test and criterion	Result
H1 (greedy advantage grows with d)	OLS slope of A_p on $\log d$, HC3, $p < 0.05$ under both optimisers	slope $+1.81/+1.52$ (grad/nsga), HC3 $p = 0.07/0.13$: not confirmed
H2a (De Ath ordering under NSGA-II)	Spearman ≥ 0.5 vs De Ath; Exploit ranks $\geq$ EI on a majority of $d \geq 6$	order $UCB < EI < Exploit$, $\rho = -0.50$, 2/4: failed
H3 (LogEI advantage shrinks under NSGA-II)	one-sided Wilcoxon on $D_p > 0$, $\alpha = 0.05$	D_p mean $+0.03$, $W = 29$, $p = 0.65$: rejected

4.6 Optimiser invariance

The gradient-free and gradient-based stacks produce closely matching relative performance on De Ath’s ten problems, though the choice of supporting statistic matters. The three-anchor ordering is identical under both optimisers, and the H3 interaction is centred at zero. The per-arm-per-problem mean log-regrets correlate at $r = 0.99$ across optimisers (50 points, Figure 4), but most of that reflects between-problem difficulty rather than ranking agreement. After removing the problem main effect the within-problem correlation is 0.93, and the within-problem arm-ranking correlation, averaged over problems, is 0.82, with the full five-arm ranking identical on 2 of 10 problems. The optimiser therefore preserves the arm ordering strongly but not perfectly. Operationally, the pattern is consistent with the gradient pathology not being the performance bottleneck and with value-underflow affecting both optimisers equally. Varying optimiser type also changes search dynamics, so the design does not isolate a single causal factor. The ordering conclusion does not rest on one metric: the mean-regret correlations and the rank-based measures (the 0.82 arm-ranking correlation and the 2-of-10 identical full rankings) support it jointly. A post-hoc random-search optimiser at the same $5000d$ budget reproduces the EI-to-LogEI ordering as well (Section 4.7), so that ordering is not an artefact of the two regimes’ evaluation budgets.

4.7 Post-hoc additions

Two arms and an interval analysis sit outside the pre-registered plan. They were declared in an append-only protocol committed before any of these cells existed. The interpretation framework, including the framings used in this section, was fixed in an amendment committed before any performance result from these cells had been read or computed; Appendix B gives the commits and timeline. Everything here is descriptive: no

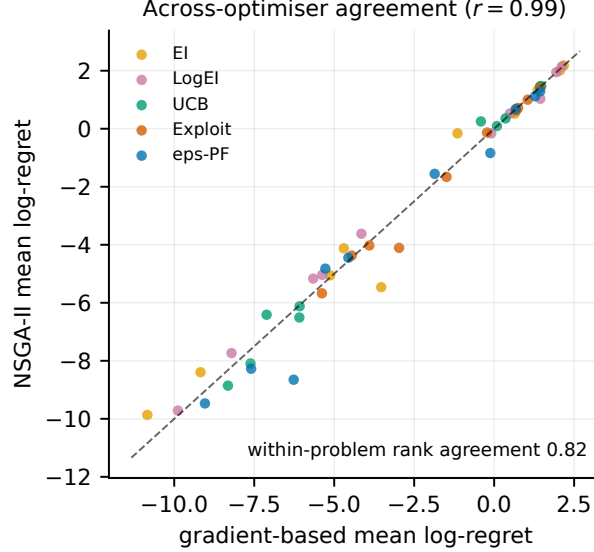


Figure 4: Absolute per-cell agreement across optimisers, over De Ath’s ten problems (50 points). Each point is one arm on one problem; axes are mean log-regret under the two optimisers. The near-diagonal correlation ($r = 0.99$) is dominated by between-problem difficulty; the within-problem agreement, which is what supports ordering-invariance, is 0.93 after removing the problem main effect and 0.82 as an arm-ranking correlation.

quantity below enters any pre-registered criterion, and the three verdicts of Table 3 are unchanged.

The exact-front eps-PF arm reproduces De Ath’s front-construction and selection mechanics line-by-line from his released code, within our surrogate stack. The residual divergences are documented in the released audit note: the NSGA-II implementation (pymoo for his pygmo), the random-number discipline, the surrogate stack (BoTorch for his GPy), and a duplicate-handling edge case. Each iteration it evolves the (mean, sigma) trade-off with NSGA-II at his settings, takes the non-dominated subset of the full evaluation archive, and selects the best-mean front member with probability 0.9, a uniform front member otherwise. The arm has 330 cells, the same paired designs and sanity gates as the main matrix, and no mechanism probe, by protocol. It is the best arm on the gradient-free side: mean log-regret -4.07 over De Ath’s ten problems, ahead of both the pre-registered variant (-3.50) and UCB (-3.24), and better than the variant on 7 of all eleven problems (Table 9). The lead over UCB is again a point estimate: the same problem-level bootstrap puts the exact-front-versus-UCB contrast at $+0.83$ with a 95% interval of $[-0.01, +1.80]$. The pre-registered variant therefore understated eps-greedy, and the corroboration of Section 4.4 is conservative. In high dimensions the exact mechanics improve on the variant (-0.52 against -0.43) but remain behind UCB (-0.71); the arm’s overall lead is driven by the low-dimensional problems.

The random-search arms optimise EI and LogEI by evaluating $5000d$ uniform samples per step, the NSGA-II arms’ exact evaluation budget with no search dynamics (660 cells). The EI-to-LogEI ordering replicates: the LogEI advantage under random search is -0.02 over De Ath’s ten problems, the same near-zero, EI-leaning gap as under both main regimes (-0.09 gradient-based, -0.12 NSGA-II, all over De Ath’s ten problems). The bounded null of Section 4.5 is therefore not an artefact of either regime’s evaluation budget or search dynamics. As a BO method, random-search acquisition optimisation is not competitive in high dimensions (mean log-regret $+0.30$ for EI against -0.06 to -0.09 under the main regimes), which is expected and separate from the ordering point. All 990 post-hoc cells pass the same gates as the main matrix (0/990 negative-regret clips; byte-identical paired designs), and the data are verifiable on the same terms as the main matrix, with archive and manifest hashes in the reproducibility statement.

5 Discussion

For practitioners. On this class of problems, the choice between raw EI and LogEI is not performance-critical, and the acquisition-optimiser type barely affects relative ranking. De Ath’s eps-greedy method remains competitive through eps-PF, and pure greedy collapses outside high dimensions. The practical ordering of concerns is therefore to include some exploration first, and to treat the precise EI variant and the acquisition-optimiser type as second-order on suites of this kind.

Ament reconciled. Our result marks out LogEI’s boundary rather than disputing the method. The gradient pathology is real, and the probe confirms it, but its translation into BO performance is bounded by the surrogate and the objective structure, which is the condition Ament et al. (2023) state in their own §6. We add pre-registered, optimiser-controlled evidence for that boundary on a suite where the LogEI gap is small.

UCB. At the fixed $\beta = 2$ used here, UCB is the strongest of the anchor arms. Under the gradient-based optimiser it is best overall, under both optimisers it is best in high dimensions, and on every high-dimensional problem it comes out ahead of EI under both optimisers (Tables 6–7). De Ath’s UCB finding is neither reproduced nor contradicted by this result. His UCB ran on the Srinivas theorem schedule for β_t , which is the setting his argument singles out as over-exploring with dimension, and a fixed- β UCB is a different arm. It does sharpen the conclusion’s scope: the data support the presence of some exploration, without ranking the mechanisms that supply it. A fixed- β UCB supplies enough exploration on this suite, as do the eps-greedy rules, and the eps-PF-versus-UCB contrast stays within problem-level noise (Appendix E).

What would change our conclusion. A powered study on a larger and more varied suite could find a robust LogEI advantage that this 11-problem suite cannot resolve; H3 is a bounded null. Two of the routes we identified were addressed post hoc (Section 4.7). Implementing De Ath’s front-construction mechanics, up to documented divergences, strengthened the eps-PF result. The budget-matched route was partially executed as a random-search arm (same budget, no search dynamics), which replicated the EI-to-LogEI ordering; a strengthened or budget-matched L-BFGS-B setting remains an open route. A dimension axis with more distinct levels could still lift H1 above its power floor, and a larger suite remains the route by which H3 could change.

Limitations. The suite is De Ath’s synthetic set, so the conclusions are about that regime and do not extend to applied BO at large. Two of our pre-registered arms differ from De Ath’s in documented ways. The gradient-based arm of eps-PF approximates the trade-off front over a candidate pool, and its NSGA-II arm evolves the front with NSGA-II over the GP, though its exploit branch and front construction differ from his eFront. Exploit is the pure-greedy limit of his eps-greedy family. Post hoc work resolved the front divergence. An exact-front arm, faithful to his released code up to documented divergences, does better than the variant overall and still sits behind UCB in high dimensions (Section 4.7). The pre-registered corroboration was therefore conservative; the Exploit reading is unchanged. The dimension axis is underpowered at eleven problems and five distinct dimensions. The value-versus-gradient question is only weakly resolvable here, because the EI-to-LogEI gap on this suite is itself small.

6 Conclusion

We show that two widely cited and apparently opposed results about Expected Improvement are compatible on the suite that produced one of them. Repairing EI’s numerics does not beat greedy, so we find no support for a broken-EI explanation of De Ath. Greedy competitiveness is real, but it is confined to high dimensions and to the eps-greedy form; pure exploitation collapses elsewhere, and the acquisition-optimiser type has little effect on relative performance. The two results coexist without contradiction, because the operative bottleneck on this suite is the presence of some exploration rather than the choice of acquisition mechanism. The frozen plan, content hashes, and analysis code are released so that the negative results can be verified directly.

Reproducibility statement

The analysis plan, hypotheses, and decision criteria were frozen and git-tagged before any production data (as defined in Section 4.1) existed (phase3-prereg, commit `fccc28a`, content-freeze commit `ea25adb`). The post-data amendment is append-only and records the analysis script’s SHA-256 (`871de0a7`) and the data archive’s hash (`d2832fab`); the post-hoc matrix of Section 4.7 is verifiable on the same terms (archive `f4a4adfa`; file-listing manifest `eb39bfd4`, recorded in the released post-hoc summary). The reproduction in Part 1 is distributional, as Section 3.1 records; seed-aggregated inference is unaffected. All figures and tables are generated from the released result artefacts by a single script, released in the public repository accompanying this paper (<https://github.com/jianxhou/exploration-or-numerics>). Initial designs are byte-identical between the paired optimisers. The analysis script’s SHA-256 is verifiable directly from the released file. The archive hashes refer to the frozen originals. The original archives, with the git provenance behind Table 5, are retained in the author’s working repository and are available on request. One limit: git timestamps are author-recorded and the chain carries no third-party anchor. The chain therefore certifies that the released plan, amendments, and analyses are mutually consistent in the disclosed order, not the wall-clock times themselves.

Compute and resources

The NSGA-II matrix (1650 cells) ran in 43.3 hours on a single consumer laptop (Apple M3, 8 GB RAM) with two worker processes. The paired gradient-based matrices add 2400 and 720 cells from earlier phases, and the post-hoc matrix of Section 4.7 adds 990 cells in 14.1 hours on the same machine. The exact-front arm averages 196 seconds per cell (median 71; the per-iteration NSGA-II front dominates), the random-search arms 51–52 seconds. The main run survived one worker out-of-memory recovery, handled by an atomic-write, skip-existing harness with no lost cells; the post-hoc run completed in a single round with no failures. The mechanism probe adds roughly 40–50 seconds per cell where it runs.

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A Full audit notes

Table 4 shows the representative Part-1 reproduction cells referenced in Section 3.1. The complete audit notes (F1, F2, F3, and F5 for De Ath, plus the separate F4 on Ament’s restart protocol), including the author-facing version reviewed by De Ath, are released in the same public repository.

B Pre-registration and amendment

The frozen pre-registration and its append-only Amendment 1 are released in full. Table 5 reconstructs the event order from the repository’s git history (tags and commits), in days relative to the Phase-3 freeze (F). Two facts matter for reading the verdicts. The gradient-based matrices and their analysis predate the freeze: when H1/H2a/H3 were frozen, the gradient-based side, including the small EI-to-LogEI gap, had been run and analysed, and no gradient-free data existed. H3 is therefore a conditional pre-registration: a directional prediction about data that did not yet exist, given the observed gradient-based side. The revision-round additions of Section 4.7 follow the same pattern, declared in an append-only protocol committed before any of the new cells existed (54fb7e1), with the interpretation framework fixed in an amendment committed before any performance result from these cells (objective value, regret, or aggregate) had been read or computed (a58f2cb). That amendment precedes the first commit of the Section-4.7 analysis script and

Table 4: Part-1 reproduction, representative of the 90/90 reproduced Table-2 cells. Median absolute regret at $T = 250$ over De Ath’s 51 published runs on Branin; our recomputation matches his published value to three significant figures for every method. In De Ath’s labels, eFront and eRandom are the eps-PF and eps-RS eps-greedy methods ($\epsilon = 0.1$) and Exploit is pure greedy, matching the arm names used in Part 2.

Method	De Ath (published)	Ours (recomputed)
LHS	1.31e-1	1.31e-1
Explore	1.66e-1	1.66e-1
EI	4.15e-6	4.15e-6
PI	2.26e-5	2.26e-5
UCB	4.42e-6	4.42e-6
PFRandom	1.67e-3	1.67e-3
eRandom	3.17e-6	3.17e-6
eFront	3.57e-6	3.57e-6
Exploit	3.08e-6	3.08e-6

output (d067002); the statement that no arm aggregation had yet been computed is a disk-state attestation recorded in the audit note, not a property of the files tracked at a58f2cb.

Per-hypothesis status at the freeze, from the same history. H2a: its target quantities (the gradient-free strategy ordering, the rank correlation with De Ath’s ordering, and the high-dimensional Exploit-versus-EI counts) did not exist in any form; the hypothesis was fully blind. H3: the gradient-side term was known more fully than a single point estimate. Under the frozen gradient-side analysis plan, the core-matrix analysis at $F-10$ (commit 8218f3e) had already found the paired EI-versus-LogEI gap not to favour LogEI and not to grow with dimension; the extension analysis re-ran that paired Wilcoxon at $T = 250$ on the full suite at $F-5$, giving the -0.08 estimate. The gradient-free term and the contrast D_p did not exist at the freeze. H3 was therefore a conditional directional prediction placed against a premise—a clear gradient-based LogEI advantage—that the pre-registered gradient-side analysis had itself already weakened, as Amendment 1 records. H1: the gradient-side trend had been computed before the freeze, in dimension-group form, under the frozen gradient-side analysis plan, with per-problem values visible in its outputs; the log-dimension regression itself was new at the freeze, but only its gradient-free half and the joint two-optimiser criterion were unevaluable then.

C Probe protocol

Two probes appear in the paper. The NSGA-II matrix carries a frozen probe that draws 4096 Sobol points per iteration with a fixed seed distinct from the BO seed, recording the value-underflow fraction (raw EI exactly zero) and the own-surface gradient-vanishing fraction. The earlier gradient-based matrices carry a lighter probe, on the EI and LogEI arms, that draws 64 Sobol points per iteration (the candidate pool that also seeds the optimiser restarts) and records the same value and gradient statistics. Value-underflow is comparable across optimisers; the gradient statistics use different sample sizes (64 versus 4096) and are compared only descriptively.

D Per-problem tables

Tables 6–9 give the per-problem mean log precise regret for all five arms under both optimisers, the per-problem H3 contrast D_p with its seed-level standard error and the LogEI-versus-EI advantage under each optimiser, and the post-hoc arms of Section 4.7. All are generated directly from the released result artefacts by the asset script.

Table 5: Event order reconstructed from git (tags/commits), in days relative to the Phase-3 pre-registration freeze F . Hashes and tags are verifiable per the reproducibility statement: file-level hashes directly from the released files, git provenance in the author’s working repository.

Event	Reference	Day	Data in existence / analyses visible
Part-1 reproduction frozen	tier1a-frozen	$F-12$	De Ath’s published runs; Table-2 gate (90/90)
Gradient analysis plan frozen	exp09-prereg	$F-12$	no Tier-2 data (plan precedes data)
Tier-2 design frozen	tier2-design-frozen	$F-11$	—
Gradient core matrix run	exp_08 (2400 cells)	$F-10$	core data created
Frozen analysis run on core	commit 8218f3e	$F-10$	gradient-side results visible
Extension matrix run	exp_10 (720 cells)	$F-6$	core + extension data
Extension analysed	commit 6c049a1	$F-5$	gradient side fully analysed
Phase-3 pre-registration frozen	phase3-prereg @ fccc28a	F	gradient side known; no NSGA-II data
NSGA-II arms + runner committed	b3ecf2d, 7065920	F	—
NSGA-II production run	exp_11 (1650 cells)	F to $F+2$	NSGA-II data created
Confirmatory analysis + Amendment 1	555a7d2, aa53ba5	$F+2$	verdicts recorded
Revision protocol committed	54fb7e1	$F+8$	before any Section-4.7 data
Framing amendment committed	a58f2cb	$F+8$	before any Section-4.7 performance result read
Section-4.7 analysis committed	d067002	$F+9$	post-hoc run complete; arm aggregation first computed

Table 6: Per-problem mean log precise regret $M_{p,a}$ under the gradient-based optimiser, as mean (SE) over the 30 seeds per cell; lower is better.

Problem	d	EI	LogEI	UCB	Exploit	eps-PF
WangFreitas	1	−1.15 (0.70)	−0.09 (0.54)	+0.09 (0.42)	−0.22 (0.66)	−1.86 (0.96)
BraninForrester	2	−9.18 (0.27)	−8.21 (0.50)	−6.09 (0.84)	−1.49 (0.64)	−7.60 (0.32)
Branin	2	−10.84 (0.27)	−9.89 (0.32)	−8.32 (0.31)	−5.38 (0.47)	−9.04 (0.26)
Cosines	2	−4.70 (0.26)	−4.15 (0.37)	−7.11 (0.55)	−3.90 (0.61)	−4.56 (0.45)
GoldsteinPriceLog	2	−3.53 (0.18)	−5.66 (0.38)	−7.61 (0.50)	−2.97 (0.90)	−6.27 (0.49)
SixHumpCamelLog	2	+1.36 (0.04)	+1.44 (0.05)	−0.41 (0.20)	+0.75 (0.27)	−0.12 (0.22)
Hartmann6Log	6	−5.13 (0.17)	−5.37 (0.25)	−6.08 (0.40)	−4.45 (0.24)	−5.28 (0.29)
Ackley8	8	+1.04 (0.03)	+1.05 (0.03)	−0.40 (0.08)	−2.08 (0.08)	−1.67 (0.10)
GSobolLog	10	+2.18 (0.04)	+2.11 (0.03)	+1.47 (0.05)	+1.05 (0.06)	+1.28 (0.06)
RosenbrockLog	10	+2.06 (0.03)	+1.95 (0.02)	+1.43 (0.03)	+1.44 (0.07)	+1.44 (0.06)
StyblinskiTangLog	10	+0.64 (0.03)	+0.50 (0.06)	+0.36 (0.10)	+0.69 (0.04)	+0.67 (0.04)

E Methods details

Objective convention and regret. The convention in this repository is maximisation, so De Ath’s minimisation problems are negated and his y_{opt} values enter Table 10 with their signs flipped as the reference optima f^* . For SixHumpCamelLog, the reference optimum is the precise value at the true minimiser, following discrepancy F1, and not the sign-flip of his stored value. Precise regret at budget T is $r = f^* - \max_t y_t$, clipped at zero (observed clips: 0/3300 paired cells), and analysed as $L = \log(\max(r, 0) + \varepsilon)$ with $\varepsilon = 10^{-12}$.

Surrogate. BoTorch **SingleTaskGP** (Matérn-5/2 ARD kernel, Gaussian likelihood at BoTorch defaults) with fixed-bounds input normalisation to $[0, 1]^d$ and outcome standardisation; hyperparameters refit every iteration by exact marginal-likelihood maximisation (**fit_gpytorch_mll**), double precision. Versions: Python 3.11.15, torch 2.12.0, BoTorch 0.17.2, GPyTorch 1.15.2, NumPy 2.4.6, pymoo 0.6.1.6.

Initial designs and pairing. Cell seed $s \in \{0, \dots, 29\}$ maps to De Ath initial-design run $s+1$; the design is injected before the first acquisition step. Ackley ($d = 8$) has no De Ath design and draws a seeded Sobol

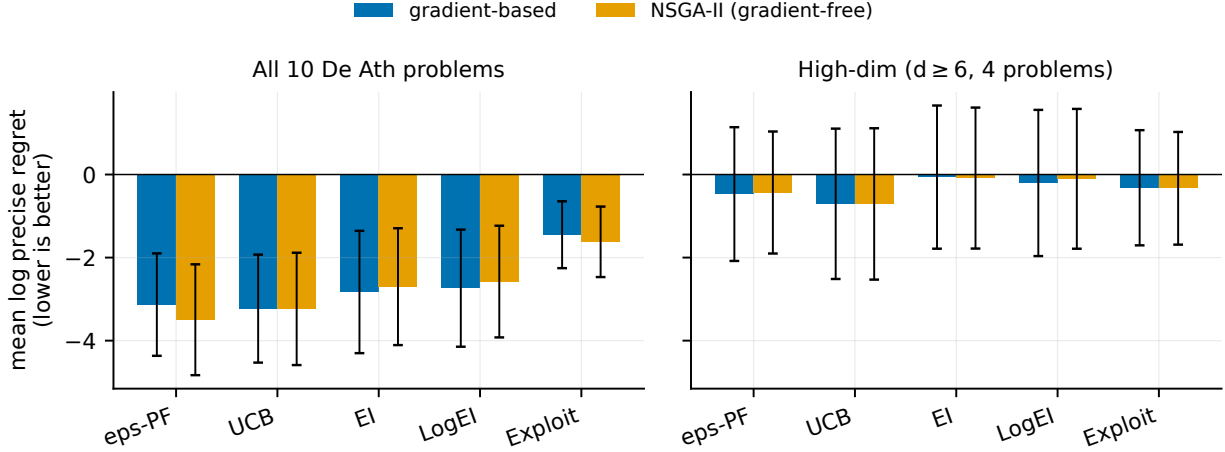


Figure 5: Shared-axis version of Figure 1: both panels on one vertical scale.

Table 7: Per-problem mean log precise regret $M_{p,a}$ under the NSGA-II optimiser, as mean (SE) over the 30 seeds per cell; lower is better.

Problem	d	EI	LogEI	UCB	Exploit	eps-PF
WangFreitas	1	-0.16 (0.60)	-0.16 (0.62)	+0.09 (0.42)	-0.13 (0.59)	-1.55 (1.27)
BraninForrester	2	-8.39 (0.55)	-7.73 (0.69)	-6.51 (0.91)	-1.66 (0.66)	-8.27 (0.35)
Branin	2	-9.87 (0.42)	-9.71 (0.36)	-8.86 (0.40)	-5.67 (0.49)	-9.47 (0.34)
Cosines	2	-4.12 (0.41)	-3.62 (0.36)	-6.41 (0.45)	-4.02 (0.68)	-4.45 (0.53)
GoldsteinPriceLog	2	-5.46 (0.28)	-5.17 (0.21)	-8.09 (0.98)	-4.11 (1.11)	-8.65 (0.64)
SixHumpCamelLog	2	+1.34 (0.12)	+1.03 (0.16)	+0.25 (0.23)	+0.72 (0.26)	-0.84 (0.36)
Hartmann6Log	6	-5.05 (0.23)	-5.04 (0.26)	-6.12 (0.45)	-4.37 (0.29)	-4.82 (0.30)
Ackley8	8	+1.05 (0.02)	+1.06 (0.02)	-0.41 (0.09)	-2.48 (0.06)	-2.19 (0.13)
GSobolLog	10	+2.18 (0.03)	+2.14 (0.03)	+1.47 (0.05)	+1.00 (0.07)	+1.12 (0.05)
RosenbrockLog	10	+2.01 (0.02)	+1.95 (0.02)	+1.46 (0.02)	+1.41 (0.07)	+1.29 (0.04)
StyblinskiTangLog	10	+0.52 (0.03)	+0.53 (0.03)	+0.36 (0.12)	+0.63 (0.05)	+0.68 (0.03)

design of size $2d$; the seed is set before the strategy is constructed, so the paired cells are byte-identical across optimisers.

Acquisitions. EI and LogEI use $\text{best}_f = \max(y)$; UCB uses $\beta = 2.0$; Exploit is the posterior mean. Under the gradient-based optimiser, the eps-PF arm ($\epsilon = 0.1$) approximates the (mean, sigma) trade-off front by taking the non-dominated max-max subset of a 64-point Sobol candidate pool at each step, and on the exploration branch it picks uniformly from that subset. The NSGA-II version instead evolves the front with a bi-objective NSGA-II over the GP, at the same settings as the other arms, and picks uniformly from the evolved non-dominated set. The exploit branch of both is the pure-greedy arm.

Failure handling. Cells are written atomically and re-run on failure by a skip-existing harness; no cell result is edited after the fact, and the matrices contain no missing cells.

Seed-level mixed models. The H1 secondary model is $L \sim C(\text{acq}, \text{Treatment}('EI')) * \text{logdim}$ with a problem-level random intercept (`groups=problem`), REML, per optimiser. The H3 secondary model is $D \sim 1$ with a problem-level random intercept, REML, on seed-level contrasts $D_{p,s}$; the reported p is one-sided in the pre-registered direction. Both are transcribed from the frozen analysis script. The released analysis output contains the H1 seed-level interaction (Exploit-versus-EI $\times \log d$: coefficient -1.81 , $p = 3.2 \times 10^{-15}$ gradient-based; -1.52 , $p = 8.4 \times 10^{-10}$ NSGA-II; negative is the direction of a growing greedy advantage). Although nominally extreme, it cannot be read as evidence for the dimension trend, and nowhere in this paper is it used that way. The reason is that $\log d$ does not vary within a problem and the model has no by-problem

Table 8: Per-problem H3 contrast D_p (LogEI advantage, gradient-based minus gradient-free), the seed-level standard error of $D_{p,s}$, and the per-problem LogEI-versus-EI advantage under each optimiser (positive = LogEI better).

Problem	d	D_p	$SE(D_{p,s})$	adv _{grad}	adv _{nsga}
WangFreitas	1	-1.06	0.62	-1.06	+0.00
BraninForrester	2	-0.31	0.87	-0.97	-0.66
Branin	2	-0.80	0.68	-0.95	-0.15
Cosines	2	-0.05	0.48	-0.55	-0.50
GoldsteinPriceLog	2	+2.42	0.48	+2.13	-0.29
SixHumpCamelLog	2	-0.38	0.24	-0.08	+0.31
Hartmann6Log	6	+0.26	0.34	+0.24	-0.02
Ackley8	8	+0.01	0.06	-0.01	-0.02
GSobolLog	10	+0.03	0.07	+0.06	+0.04
RosenbrockLog	10	+0.05	0.04	+0.11	+0.06
StyblinskiTangLog	10	+0.15	0.07	+0.15	-0.01

Table 9: Per-problem mean log precise regret for the post-hoc arms (Section 4.7), as mean (SE) over the 30 seeds per cell, beside the pre-registered eps-PF variant under NSGA-II.

Problem	d	eps-PF (exact front)	EI (RS)	LogEI (RS)	eps-PF (variant)
WangFreitas	1	-0.26 (0.69)	-0.57 (0.88)	+0.01 (0.48)	-1.55 (1.27)
BraninForrester	2	-10.47 (0.26)	-9.58 (0.52)	-9.65 (0.50)	-8.27 (0.35)
Branin	2	-10.58 (0.27)	-10.20 (0.24)	-10.47 (0.24)	-9.47 (0.34)
Cosines	2	-5.56 (0.64)	-4.41 (0.31)	-4.48 (0.38)	-4.45 (0.53)
GoldsteinPriceLog	2	-10.79 (0.66)	-7.12 (0.30)	-6.99 (0.29)	-8.65 (0.64)
SixHumpCamelLog	2	-0.97 (0.30)	+1.38 (0.05)	+1.34 (0.09)	-0.84 (0.36)
Hartmann6Log	6	-5.46 (0.25)	-4.12 (0.06)	-4.09 (0.08)	-4.82 (0.30)
Ackley8	8	-0.57 (0.05)	+1.25 (0.01)	+1.22 (0.02)	-2.19 (0.13)
GSobolLog	10	+1.40 (0.02)	+2.28 (0.02)	+2.28 (0.02)	+1.12 (0.05)
RosenbrockLog	10	+1.57 (0.02)	+2.16 (0.01)	+2.14 (0.01)	+1.29 (0.04)
StyblinskiTangLog	10	+0.43 (0.06)	+0.87 (0.01)	+0.85 (0.01)	+0.68 (0.03)

random slope, so the interaction is tested against seed-level residual variance, and that overstates the effective sample for a between-problem effect. The problem-level regression is the primary estimate throughout.

Post-hoc bootstrap. Percentile bootstrap over problems (resampling problems with replacement, problem-level means held fixed), $B = 10,000$, 95% intervals, fixed seed, declared in the revision protocol. The point estimates and intervals are as follows. Over De Ath’s ten problems, the LogEI advantage is $-0.12 [-0.30, +0.04]$ under NSGA-II and $-0.09 [-0.58, +0.50]$ under the gradient-based optimiser, where one low-dimensional problem, GoldsteinPriceLog at $+2.13$, widens the gradient interval. Over all eleven problems the values are $-0.11 [-0.28, +0.03]$ under NSGA-II and $-0.08 [-0.53, +0.46]$ under the gradient-based optimiser, and the mean D_p is $+0.03 [-0.39, +0.57]$. Descriptive only. Two further contrasts computed post hoc with the same machinery, quoted in Sections 4.4 and 4.7: eps-PF-versus-UCB under NSGA-II over De Ath’s ten problems $+0.26 [-0.46, +0.93]$, and exact-front-versus-UCB $+0.83 [-0.01, +1.80]$. The Section-4.5 leave-one-out check (dropping GoldsteinPriceLog: mean $D_p = -0.21$, one-sided Wilcoxon $p = 0.84$) is likewise post hoc, on the same problem-level values.

Table 10: The eleven problems: dimension, the De Ath problem it maps to (Ackley has none; it was added at pre-registration), and the reference optimum f^* used for precise regret (this repo maximises; log-prefixed De Ath problems are his log-transformed variants).

Problem	d	De Ath problem	f^*
WangFreitas	1	WangFreitas	+4
BraninForrester	2	BraninForrester	+16.644
Branin	2	Branin	-0.397887
Cosines	2	Cosines	+1.6
GoldsteinPriceLog	2	logGoldsteinPrice	-1.09861
SixHumpCamelLog	2	logSixHumpCamel	+9.54516
Hartmann6Log	6	logHartmann6	+1.20068
Ackley8	8	—	+0
GSobolLog	10	logGSobol	+6.93147
RosenbrockLog	10	logRosenbrock	+0.693147
StyblinskiTangLog	10	logStyblinskiTang	-2.12086